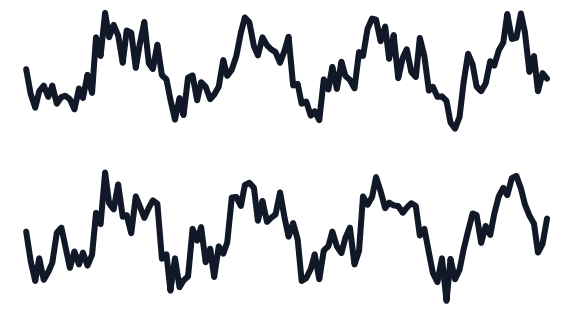


CoG-SAGE: Color-Order Guided, Session-Aware Gated EEG Decoding

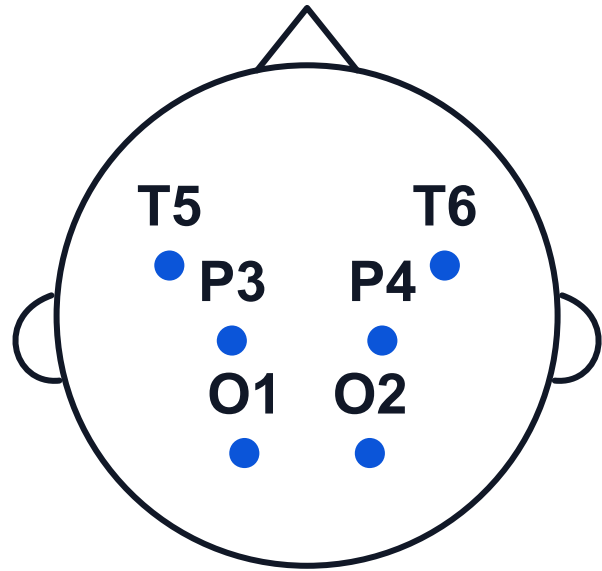
(a) EEG Input

Raw EEG epochs

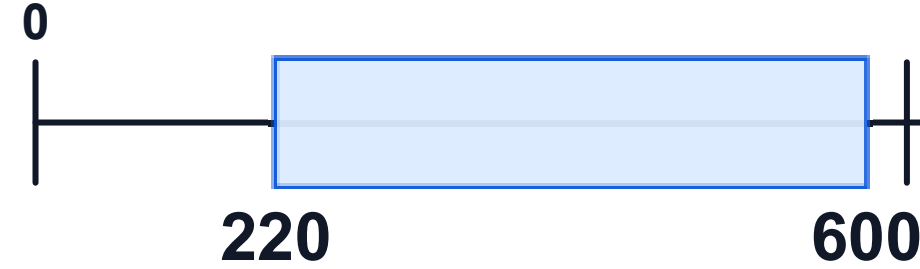


$$X \in \mathbb{R}^{B \times 6 \times 380}$$

Posterior 6 channels



Samples 220:600
($\approx 20-400$ ms post-stimulus)



7-class labels



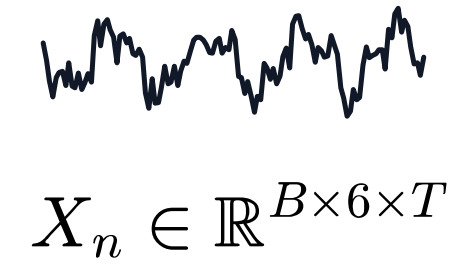
Session-stratified split



(b) Preprocessing & Auxiliary Features

1) Raw normalization (train-only)

Train-only normalization of raw input
(z-score, fit on training split only)

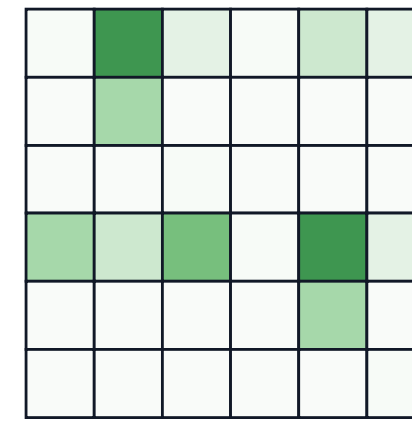


$$X_n \in \mathbb{R}^{B \times 6 \times T}$$

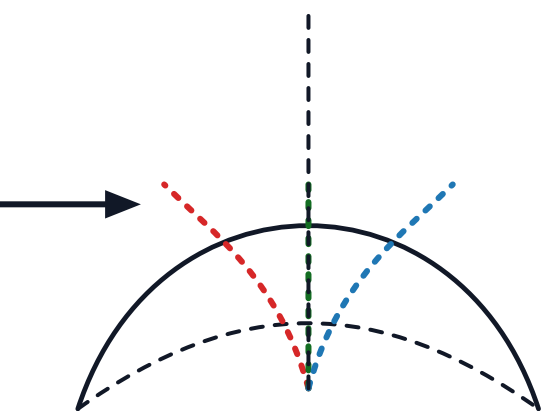


2) Auxiliary feature branch

(i) Covariance / Riemann
auxiliary features



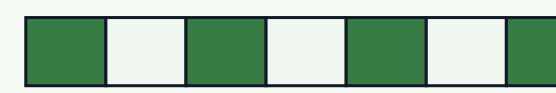
Covariance
(6×6)



Log-Euclidean /
Riemann mapping

(ii) Train-only normalization of auxiliary features

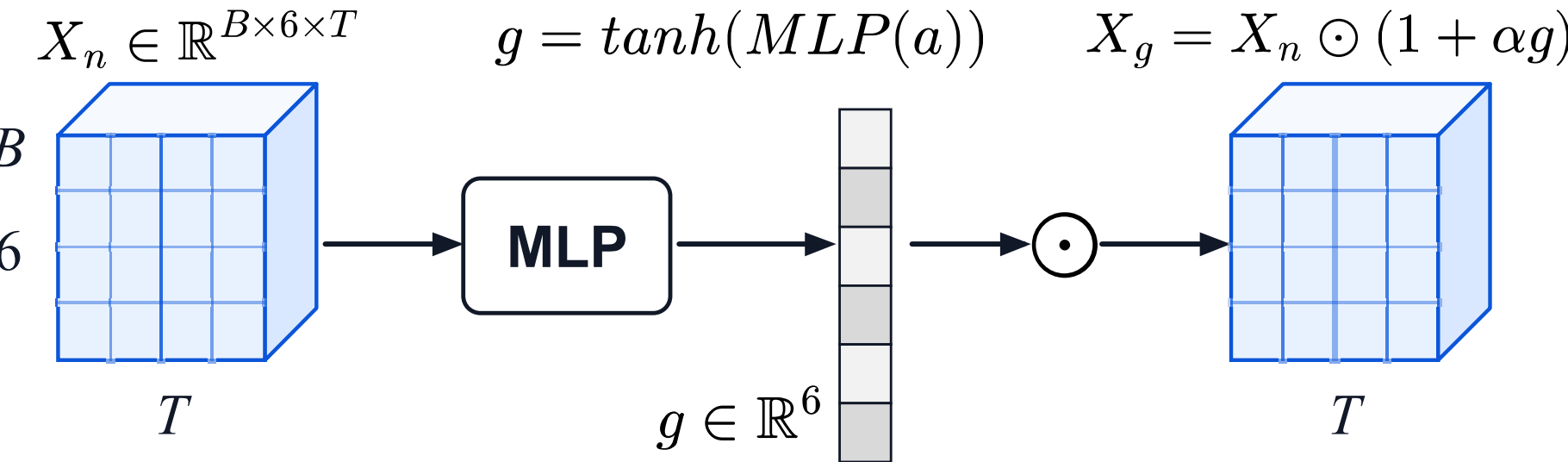
aux z-score,
fit on training split only



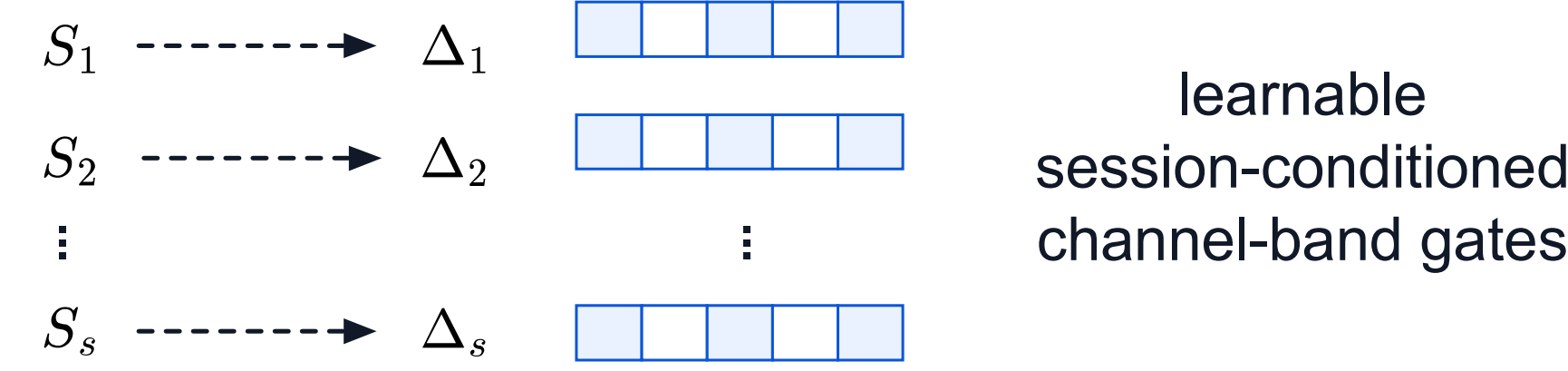
$$a \in \mathbb{R}^{d_{aux}}$$

(c) Representation Learning

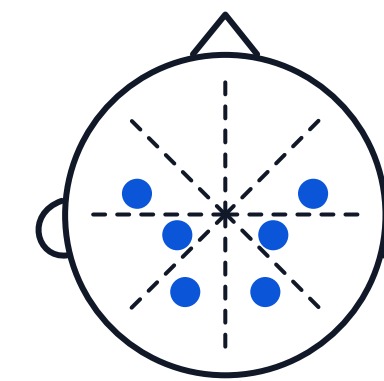
C1 Auxiliary-guided channel/spatial gating



C2 Session-aware SRM (channel-band spectral modulation)

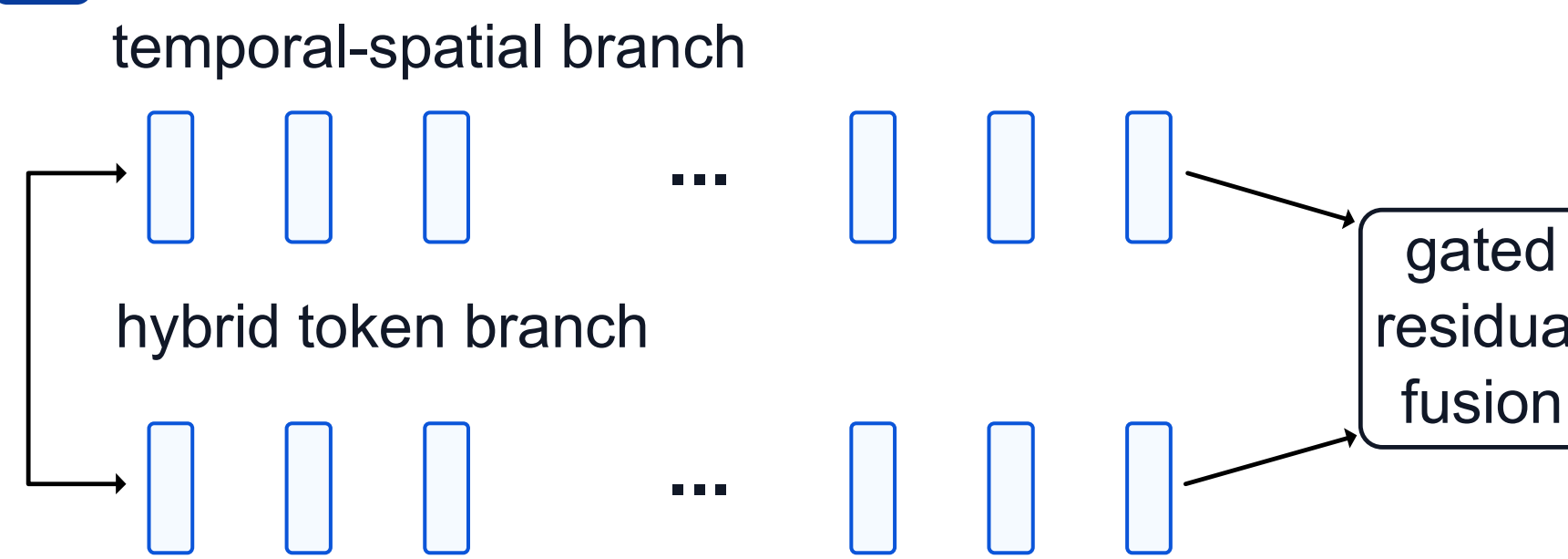


C3 Session-adaptive CAR front-end



learnable
reference scale

C4 Hybrid raw backbone



C5 Fused representation

$$z \in \mathbb{R}^{d_z}$$



(d) Prediction Heads

Main classifier –
7-class logits



HSV prototype head –
prototype similarities / logits
7-class outputs



Gated adjacent
class-query head –
residual 7-class logits



Feature-state residual head –
residual 7-class logits



Combined logits

$$l = l_{main} + \lambda_{HSV} l_{HSV} + \lambda_{CQ} l_{CQ} + \lambda_{FT} l_{FT}$$

(e) Training Objective

Loss components

Adjacent Soft Focal Loss
on combined logits

Center Loss on z

HSV Prototype
Logit Regularization
on prototype logits

Total objective

$$L = L_{ASF}(l, y) + \lambda_c L_{center}(z, y) + \lambda_p L_{proto}(l_{HSV}, y)$$

Training strategies:

Mixup, EMA, Checkpoint averaging

Raw EEG flow

Auxiliary feature flow

Session-conditioning flow

Loss / gradient flow